Fake data: variety, physics and the 156 sensors split by stage

SOFC plant · 15 October 2026 · Contents: 1 The answer · 2 Data split by stage · 3 Do the 106 follow the 50? · 4 Order of work

1 · The answer

Variety and physics models can be used. They are layers 3 and 4 of the agreed plan, and only layers 1–2 are built so far. Before we change the 50 stage sensors, we need to know how the other 106 depend on them in each stage. Otherwise the fake 50 and the copied 106 stop matching. That split is done now.

1	**Stage timeline**	BUILT
2	**Fill with real pieces**	BUILT
3	**Add variety (on the 50)**	NEXT
4	**Physics models (on the 50)**	NEXT
5	**Rebuild the other 106 per stage**	NEW STEP
6	**Checks**	NEXT

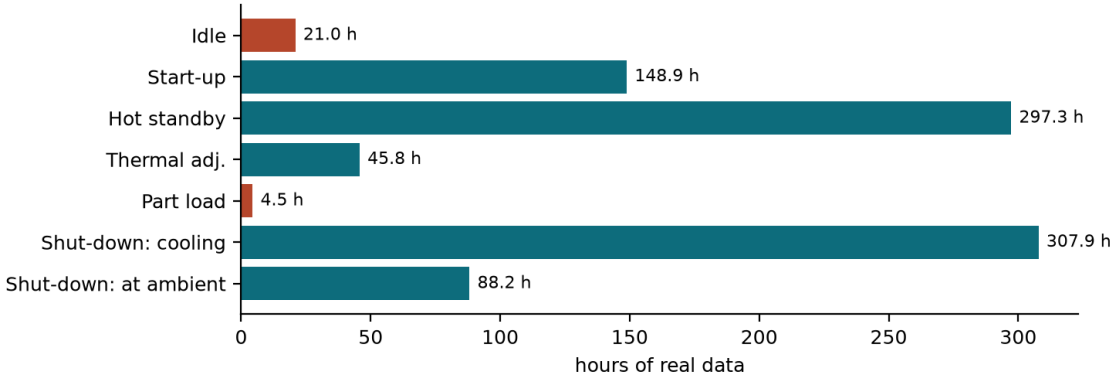
Why they were not the first step. (1) They break the match: today a fake hour is a copied real hour, so all sensors agree; changing the 50 leaves the 106 behind. (2) Physics covers few signals: stack heat-up ramp (≈ 60 K/h heater, ≈ 50 °C/h stack), cool-down curve (≈ 21 h, 5–9 h at high air), voltage–current per string; no such laws for supply pressures, reformer, heater currents or safety monitors. (3) Some stages are tiny: Part load 4.5 h, Idle one day. (4) Copying had to work first: correct stage order, gas phases and joins were finished this week.

Finding	Number
Sensors split into 7 stage files (50 stage-rule + 106 others)	156
Other sensors computable from the 50 in at least one stage	51
Other sensors with their own behaviour in most stages	56
Sensors the current filler does not copy at all	42

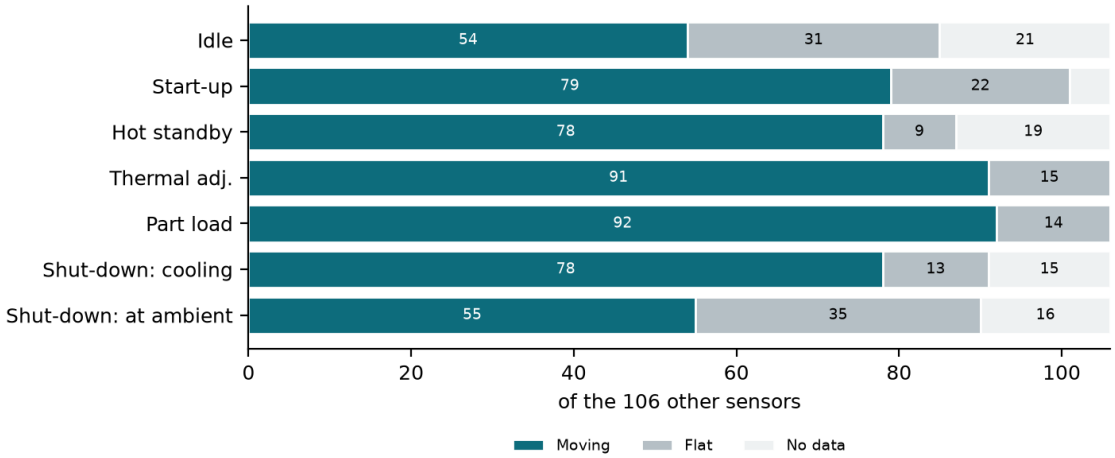
If you remember one thing: variety and physics change only the 50; the 106 must then be rebuilt stage by stage, and today's split gives the recipe.

2 · The 156 sensors, split into 7 stage files

Every minute already has a stage label. The 156 kept sensors were cut along those labels, one file per stage; Shut-down is split into Cooling and At ambient.



Flat = uses less than 3 % of the sensor's normal range in that stage; No data = under 30 % filled.



Stage	Hours	Stretches	Days	Moving	Flat	No data
Idle	21.0	2	1	54	31	21
Start-up	148.9	14	17	79	22	5
Hot standby	297.3	26	23	78	9	19
Thermal adj.	45.8	18	12	91	15	0
Part load	4.5	10	5	92	14	0
Shut-down: cooling	307.9	14	23	78	13	15
Shut-down: at ambient	88.2	8	8	55	35	16

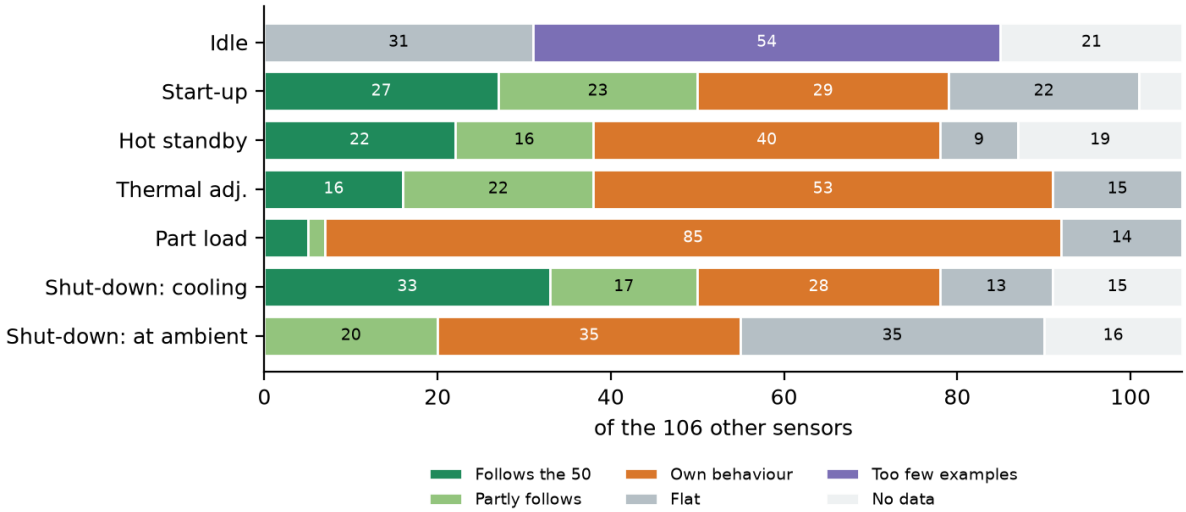
Found on the way: 42 sensors are never copied into the fake data today (string power, set-points, ramp and on/off EIRC 4,s,3/4/5/7/8 = 23; stack totals EIR 4,s,12/15/18 = 15; heater currents = 3; QIR 1,4 = 1). They are in the new stage files.

Files: data_analysis/sensors_by_stage/data/1_Idle.parquet … 7_Shut-down_At_ambient.parquet, sensor_map.csv, stage_stats.csv; notebook data_analysis/sensors_by_stage.ipynb

If you remember one thing: each stage has its own file with all 156 sensors, and part of the 106 hardly moves in each stage, so those need no model.

3 · Can the 106 other sensors be computed from the 50?

Per stage and moving sensor, a straight-line and a curve model predict the sensor from the 50 stage sensors. Each is scored on whole runs it never saw (R^2: follows ≥ 0.9, partly 0.6–0.9, own < 0.6).



Stage	Reading
Start-up	Half can be computed (27 follow, 23 partly).
Shut-down: cooling	Best stage: 33 follow, 17 partly.
Hot standby	22 follow; 40 own (hold level, supply pressures, reformer settings vary by run).
Thermal adjustment	Mostly own (53): short stretches, each different.
Part load	85 look 'own', but only 4.5 h exist; the test mainly says 'not enough runs'.
Shut-down: at ambient	Plant off: flat (35) or slow drift.
Idle	One day: cannot be tested.

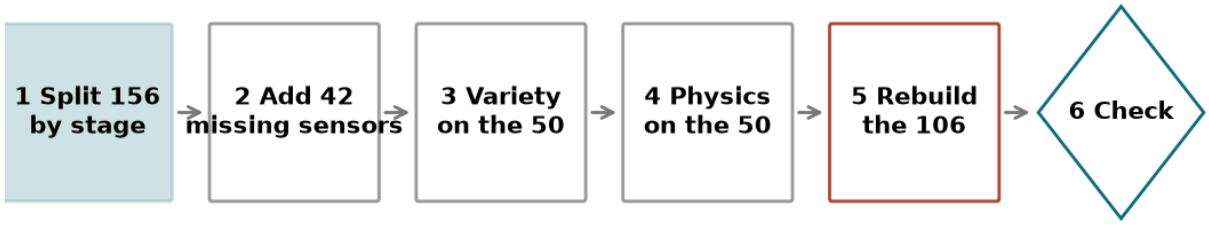
Across all stages: 5 always follow (string power EIRC 4,s,3 = voltage × current), 10 follow or partly, 35 mixed, 56 mostly own. One model per sensor will not work; the recipe is chosen per stage and sensor (other_sensors_by_stage.csv, 742 pairs).

Class	Recipe for the fake data
Follows the 50	Compute from the varied/physics-made 50 with that stage's model
Partly follows	Stage model + real wiggle copied from a real piece
Own behaviour	Copy a real piece of the same stage and gas phase
Flat	Stage level + real noise
Too few examples	Copy real
No data	Leave blank and flag

If you remember one thing: about a third of the other sensors can be computed from the 50 in heat-up and cool-down; in the hot stages most cannot, so we keep copying real pieces there.

4 · Order of work

Your order is right: split first, then make the 50 with variety and physics, then the 106. One small fix comes first so all 156 reach the fake data.



Step	What	Uses
2 Missing sensors	Add the 42 sensors to the filler's copy cache	stage files
3 Variety	Real-sized noise, stretch heat-up ± 10–20 %, shift a hold level, reshuffle wiggles; stay inside the stage's real band	stage_stats.csv
4 Physics	Heater ramp ≈ 60 K/h to 300 °C, stack ≈ 50 °C/h; cool-down ≈ 21 h (5–9 h at high air); V–I line per string, ≤ 16 A	stage notebooks
5 Rebuild the 106	Recipe per stage and sensor from section 3	other_sensors_by_stage.csv
6 Check	50-sensor stage checker ≥ 80 % per block; values inside real bands	stage checker

Two choices for you. A: for 'own behaviour' sensors keep copying real pieces (proposed) or use a simple drift model. B: in Part load copy real pieces for the 106 and use physics only for current, voltage and power (proposed).

If you remember one thing: a small fix (42 missing sensors), then variety and physics on the 50, then the 106 rebuilt from the per-stage recipe.